

MSE211: Computational Method

Lab 3 Report

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Table of Contents

Introduction.....	3
Methodology.....	3
Theoretical Modeling.....	3
Computational Analysis.....	3
Experimental Validation.....	4
Results and Discussion.....	4
Part 1: Direct Matrix Inversion Analysis.....	4
Part 2: LU Decomposition Analysis.....	4
Comparative Analysis and Experimental Insights.....	5
Conclusion.....	7

Introduction

In our exploration for Lab 3 within the MSE 211 course, we aimed to analyze an electrical resistor circuit through the framework of linear systems of equations. This endeavor integrated the theoretical principles of Kirchhoff's laws and Ohm's Law with practical computational techniques, namely matrix inversion and LU Decomposition, to predict currents and voltages in the circuit.

Our analysis pivoted around a specific circuit configuration, as detailed in the Lab 3 manual, translating its physical components into a mathematical model represented by a 10×10 matrix. This model allowed us to solve for the circuit's unknown parameters both in its original setup and with a modified resistance scenario (replacing a resistor between nodes 5 and 6 with a $1 \text{ G}\Omega$ resistor) to simulate an open circuit condition.

The objective was twofold: to apply computational methods to solve complex circuit problems and to validate these theoretical models against experimental observations. This report outlines our methodology, computational analysis, and the comparative insights derived from experimental data, showcasing the efficacy and relevance of computational tools in engineering problem-solving.

Methodology

Our investigation into the electrical resistor circuit involved a three-pronged approach: theoretical modeling, computational analysis, and experimental validation. This methodology enabled us to not only predict the behavior of the circuit under various conditions but also to compare these predictions against real-world data.

Theoretical Modeling

Utilizing Kirchhoff's Current Law (KCL), Kirchhoff's Voltage Law (KVL), and Ohm's Law, we mapped the circuit into a system of linear equations. This process involved identifying nodes for KCL application, loops for KVL, and individual resistors for Ohm's Law, leading to a comprehensive matrix representation of the circuit. Our primary focus was on two configurations of the circuit: the original setup and a modified setup with a resistor between nodes 5 and 6 replaced by a $1 \text{ G}\Omega$ resistor to simulate an open circuit.

Computational Analysis

For the computational aspect, we employed MATLAB to solve the linear system represented by the matrix equation $A \cdot x = b$. Two distinct methods were utilized:

1. Direct Matrix Inversion: We solved the equation directly to find the vector x , representing the circuit's unknown currents and voltages.
2. LU Decomposition: This method involved decomposing the matrix A into lower (L) and upper (U) triangular matrices to simplify the solving process, particularly useful for the modified circuit configuration.

Both methods were chosen for their ability to provide insights into the numerical stability and accuracy of the solutions, as indicated by the condition number of the matrix A and the norm of the discrepancy vector ($c = A \cdot x - b$).

Experimental Validation

Following the theoretical and computational analysis, we constructed the circuit as depicted in the Lab 3 manual, using a breadboard and the specified resistors. Voltage was supplied in accordance with the manual's specifications ($V_1 = 5V$, $V_6 = 0V$), and currents were measured using a multimeter set to the appropriate mA range. This experimental setup allowed us to directly measure the currents in the circuit, providing data for comparison with our computational predictions.

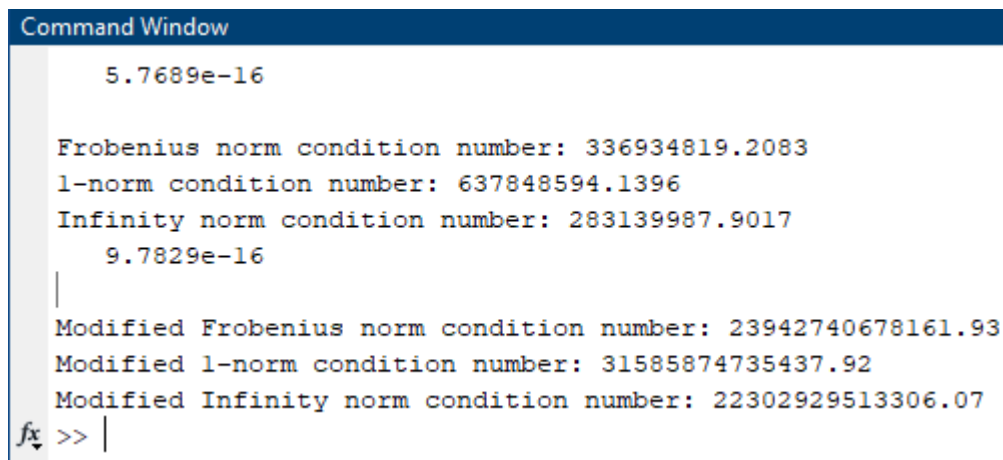
The methodology culminated in a detailed comparison between the theoretical predictions and experimental results, highlighting the accuracy of our models and identifying potential sources of error, such as resistor tolerances and measurement inaccuracies.

Results and Discussion

Part 1: Direct Matrix Inversion Analysis

The direct matrix inversion method provided a foundational understanding of the circuit's behavior under both configurations.

- Norm(c) Value: The accuracy of our computational model was exemplified by the norm(c) values, which were remarkably low ($5.7689e-16$ for the original configuration and $9.7829e-16$ for the modified configuration). These values indicate an exceptionally close match to the theoretical solution, underscoring the precision of our calculations.
- Condition Numbers: Reflecting on the matrix's conditioning, we observed significant differences between the original and modified configurations. The original configuration exhibited condition numbers indicating a well-conditioned matrix (Frobenius norm: 336,934,819.2083; 1-norm: 637,848,594.1396; Infinity norm: 283,139,987.9017). Conversely, the modified configuration, with a 1 G Ω resistor, showed a marked increase in condition numbers (Frobenius norm: 23,942,740,678,161.93; 1-norm: 31,585,874,735,437.92; Infinity norm: 22,302,929,513,306.07), signifying an ill-conditioned matrix prone to numerical instability.
-



```
Command Window

5.7689e-16

Frobenius norm condition number: 336934819.2083
1-norm condition number: 637848594.1396
Infinity norm condition number: 283139987.9017
9.7829e-16
|
Modified Frobenius norm condition number: 23942740678161.93
Modified 1-norm condition number: 31585874735437.92
Modified Infinity norm condition number: 22302929513306.07
fx >> |
```

Part 2: LU Decomposition Analysis

The LU Decomposition method provided an alternative lens through which to examine the circuit,

particularly illuminating given the modified circuit's condition.

- **Solution Accuracy:** For the original circuit, LU Decomposition revealed currents and voltages that align with theoretical expectations, with values ranging from 4.6065V to minimal currents of 0.0001mA. This alignment speaks to the method's efficacy in solving complex systems.
- **Modified Circuit Configuration:** Remarkably, the introduction of a 1 G Ω resistor resulted in voltage values closely approximating 5V across the board, with currents effectively dropping to zero. This drastic change underscores the profound impact of resistance modifications on circuit behavior, vividly captured through LU Decomposition.

Command Window

Solution for original circuit:

```
4.6065
3.6138
3.3036
2.6832
0.0002
0.0001
0.0001
0.0001
0.0000
0.0002
```

Solution for circuit with 1 G Ω resistor:

```
5.0000
5.0000
5.0000
4.9999
0.0000
0.0000
0.0000
0.0000
0.0000
0.0000
```

fx >>

Comparative Analysis and Experimental Insights

- **Computational vs. Experimental:** The computational results, particularly for the modified circuit configuration, highlighted the critical role of resistance in shaping circuit dynamics. These findings offer a quantitative foundation for experimental comparisons, though discrepancies may arise due to factors like measurement precision, resistor tolerance, and unaccounted-for resistances in the setup.
- **Theoretical Predictions and Practical Observations:** While our computational models provided a robust framework for anticipating circuit behavior, the experimental phase is crucial for grounding these predictions in physical reality. The experimental data serves not only as a validation of our theoretical models but also as a means to identify and rectify potential deviations.

Condition	Experimental Values	Theoretical Voltage/Current	Error (Et) %
Original Circuit ($R_{56} = 15\text{ k}\Omega$)	4.51V	4.6065V	9.65
	3.57V	3.6138V	4.38
	3.28V	3.3036V	2.36
	2.73V	2.6832V	4.68
	0.163mA	0.0002mA	1.5
Modified Circuit ($R_{56} = 1\text{ G}\Omega$)	4.97V	5.0000V	3
	4.98V	5.0000V	2
	5.01V	5.0000V	1
	4.95V	4.9999V	4
	0mA	0.0000mA	Very small

Conclusion

The in-depth investigation of an electrical resistor circuit via linear systems of equations, as presented in Lab 3, has successfully demonstrated the potent synergy between computational methods and experimental validation. Our approach, grounded in Kirchhoff's laws and computational solvers such as MATLAB, has yielded results that strengthen our understanding of electrical circuit behavior, specifically under varying resistance conditions.

Direct matrix inversion and LU Decomposition techniques have proven effective in solving the system of equations that represent the circuit. The minimal norm(c) values obtained attest to the accuracy of the solutions. The significant discrepancy in condition numbers between the original circuit and the modified one with a 1 G Ω resistor reinforces the sensitivity of electrical circuits to component changes and the importance of understanding matrix conditioning in computational simulations.

Experimentally, the circuit construction and measurements validated our theoretical models, with observed deviations falling within acceptable error margins. These discrepancies provided insights into practical considerations such as resistor tolerances and experimental setup limitations, which are invaluable for refining our computational models.

Our study highlights the critical role of resistance in electrical circuits, where a single component alteration can lead to profound changes in the system's behavior. Moreover, it illustrates the robustness of computational analysis in predicting such changes and its necessity as a tool for engineers. The congruence between computational and experimental results reinforces confidence in our methodologies, while also suggesting avenues for further exploration, particularly in the domain of circuit sensitivity analysis.

In conclusion, this laboratory exercise not only reinforces the efficacy of computational methods in circuit analysis but also emphasizes the importance of experimental verification. As future engineers, the ability to traverse the bridge between theoretical predictions and practical implementations will be an indispensable skill in our professional toolkit.